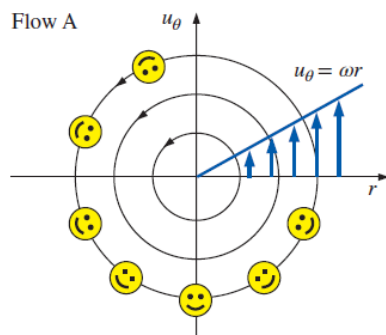


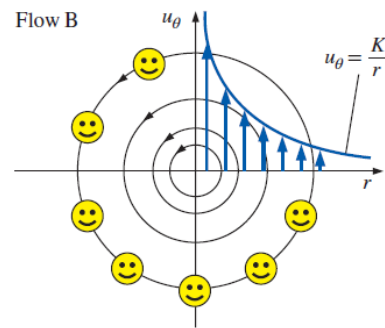
Irrotational and Rotational Flow

Vorticity and Rotationality:

- If the vorticity at a point in a flow field is nonzero, the fluid particle that happens to occupy that point in space is rotating; the flow in that region is called **rotational**.
- If the vorticity in a region of the flow is zero (or negligibly small), fluid particles there are not rotating; the flow in that region is called **irrotational**.
- The rotation of fluid elements is associated with wakes, boundary layers, flow through turbomachinery (fans, turbines, compressors, etc.), and flow with heat transfer.
- The vorticity of a fluid element cannot change except through the action of viscosity, nonuniform heating (temperature gradients), or other non-uniform phenomena.

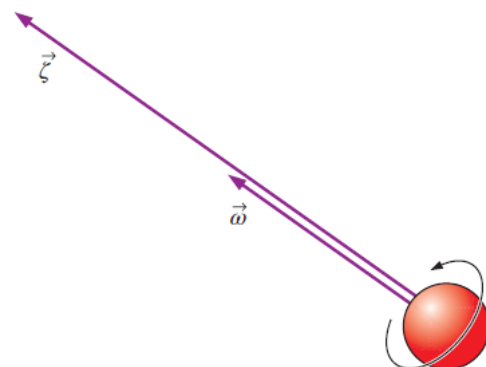
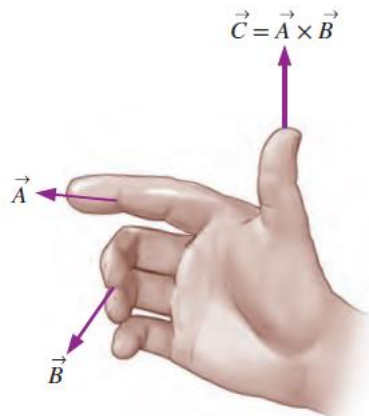


Flow A is **rotational**. Physically, this means that individual fluid particles rotate as they revolve around the origin.



Flow B is **irrotational**. Physically, fluid particles do *not* rotate as they revolve in circles about the origin.

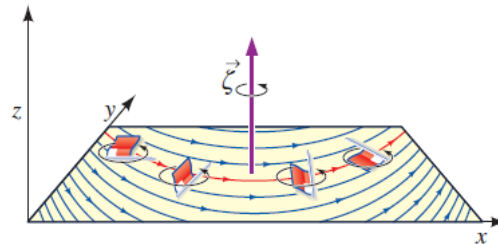
A simple analogy can be made between flow A and a merry-go-round or roundabout, and flow B and a Ferris wheel. As children revolve around a roundabout, they also rotate at the same angular velocity as that of the ride itself. This is analogous to a rotational flow. In contrast, children on a Ferris wheel always remain oriented in an upright position as they trace out their circular path. This is analogous to an irrotational flow.



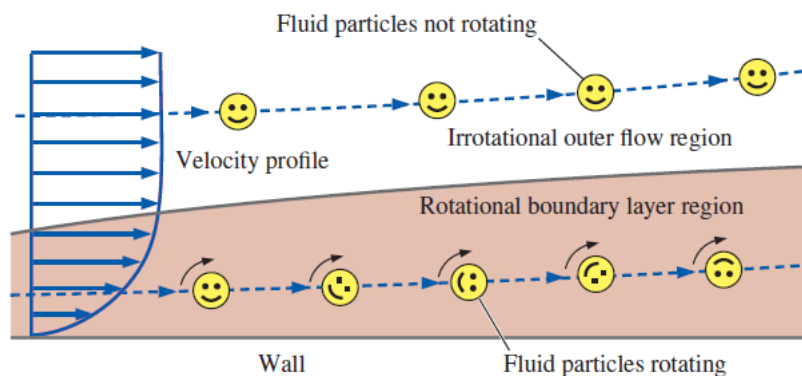
The direction of a vector cross-product is determined by the **right-hand rule**.

The **vorticity vector** is equal to **twice** the **angular velocity** vector of a rotating fluid particle.

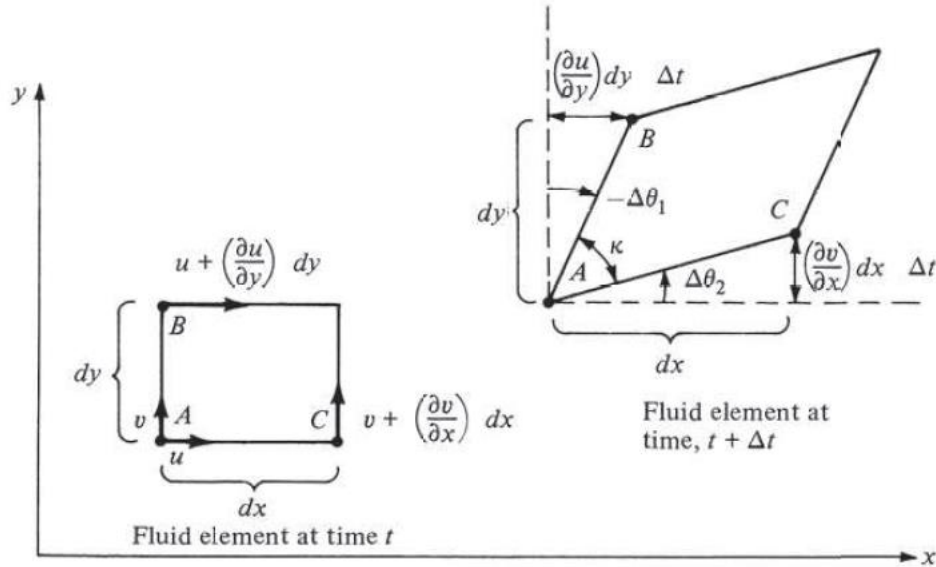
For two-dimensional flow in the xy -plane, the vorticity vector always points in the z - or $-z$ -direction. In this illustration, the flag-shaped fluid particle rotates in the counter clockwise direction as it moves in the xy -plane; its vorticity points in the positive z -direction as shown.



The difference between rotational and irrotational flow: fluid elements in a rotational region of the flow rotate, but those in an irrotational region of the flow do not. Note that the fluid elements would also *distort* as they move along in the flow.



- Consider an infinitesimal fluid element moving in a flow field.
- As it translates along a streamline, it may also rotate and in addition, its shape may become distorted (due to shear stress).
- The amount of rotation and distortion depends on the velocity field.
- Fluid element rotates without distortion – **Solid Body Rotation**.
- The measure of flow rotation at a point is **Vorticity**.
- **Counter-clockwise rotations** by convention are considered **positive**.



From the figure,

- Velocity at point A along x-direction = u
- Velocity at point A along y-direction = v
- Velocity at point B along x-direction = $u + \left(\frac{\partial u}{\partial y}\right) dy$
- Velocity at point C along y-direction = $v + \left(\frac{\partial v}{\partial x}\right) dx$

Distance in y-direction that point A moves during time increment $\Delta t = v\Delta t$

Distance in y-direction that point C moves during time increment $\Delta t = \left(v + \left(\frac{\partial v}{\partial x}\right) dx\right) \Delta t$

Net displacement in the y-direction of C relative to A = $\left(v + \left(\frac{\partial v}{\partial x}\right) dx\right) \Delta t - v\Delta t = \left(\frac{\partial v}{\partial x}\right) dx \Delta t$

Similarly, Net displacement in the x-direction of B relative to A = $\left(\frac{\partial u}{\partial y}\right) dy \Delta t$

From the geometry,

$$\tan \Delta\theta_2 = \frac{\left[\left(\frac{\partial v}{\partial x}\right) dx\right] \Delta t}{dx} = \frac{\partial v}{\partial x} \Delta t$$

Since $\Delta\theta_2$ is a small angle, $\tan \Delta\theta_2 \approx \Delta\theta_2$. Hence,

$$\Delta\theta_2 = \frac{\partial v}{\partial x} \Delta t$$

Similarly,

$$\Delta\theta_1 = -\frac{\partial u}{\partial y} \Delta t$$

Consider the angular velocities of lines AB and AC

$$\frac{d\theta_1}{dt} = \lim_{\Delta t \rightarrow 0} \frac{\Delta\theta_1}{\Delta t} = -\frac{\partial u}{\partial y} \quad \text{and} \quad \frac{d\theta_2}{dt} = \lim_{\Delta t \rightarrow 0} \frac{\Delta\theta_2}{\Delta t} = \frac{\partial v}{\partial x}$$

By definition, the angular velocity of the fluid element as seen in the xy plane is the average of the angular velocities of lines AB and AC .

$$\omega_z = \frac{1}{2} \left(\frac{d\theta_1}{dt} + \frac{d\theta_2}{dt} \right) = \frac{1}{2} \left(-\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x} \right) = \frac{1}{2} \left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \right)$$

In three-dimensional space, the x and y components of $\boldsymbol{\omega}$ can be obtained in a similar fashion. The resulting angular velocity of the fluid element in three-dimensional space is

$$\begin{aligned} \boldsymbol{\omega} &= \omega_x \hat{i} + \omega_y \hat{j} + \omega_z \hat{k} \\ \boldsymbol{\omega} &= \frac{1}{2} \left[\left(\frac{\partial w}{\partial y} - \frac{\partial v}{\partial z} \right) \hat{i} + \left(\frac{\partial u}{\partial z} - \frac{\partial w}{\partial x} \right) \hat{j} + \left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \right) \hat{k} \right] \end{aligned}$$

The angular velocity of a fluid element plays an important role in theoretical aerodynamics. we define a new quantity, **vorticity**, which is simply twice the angular velocity.

$$\boldsymbol{\xi} \equiv 2\boldsymbol{\omega}$$

$$\boldsymbol{\xi} = \left[\left(\frac{\partial w}{\partial y} - \frac{\partial v}{\partial z} \right) \hat{i} + \left(\frac{\partial u}{\partial z} - \frac{\partial w}{\partial x} \right) \hat{j} + \left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \right) \hat{k} \right]$$

From a vector identity, i.e. **Curl of a velocity vector**

$$\nabla \times \vec{V} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ u & v & w \end{vmatrix} = \left(\frac{\partial w}{\partial y} - \frac{\partial v}{\partial z} \right) \hat{i} + \left(\frac{\partial u}{\partial z} - \frac{\partial w}{\partial x} \right) \hat{j} + \left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \right) \hat{k}$$

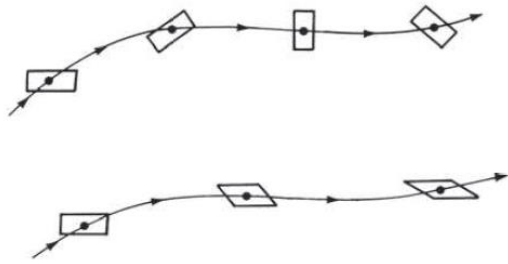
Hence, we have the important result that

$$\boldsymbol{\xi} = \nabla \times \vec{V}$$

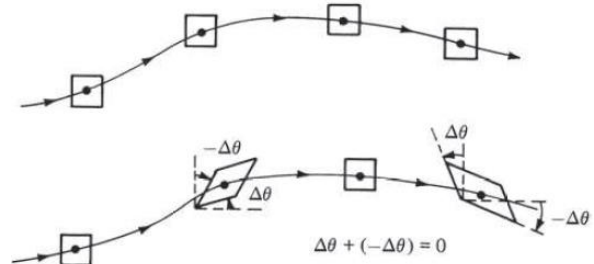
In a velocity field, the curl of the velocity is equal to the vorticity.

The above condition leads to two important definitions:

1. If $\nabla \times \vec{V} \neq 0$ at every point in a flow, the flow is called **rotational**. This implies that the fluid elements have a finite angular velocity.
2. If $\nabla \times \vec{V} = 0$ at every point in a flow, the flow is called **irrotational**. This implies that the fluid elements have no angular velocity; rather, their motion through space is a pure translation.



Fluid elements in a rotational flow



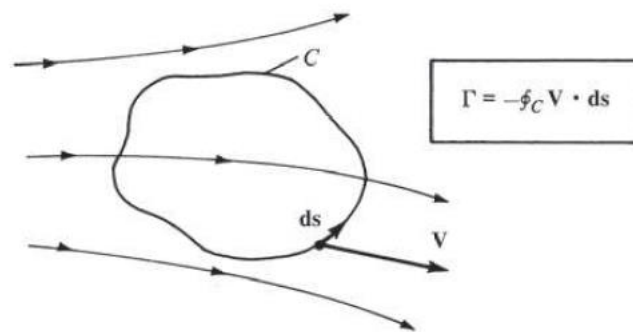
Fluid elements in a irrotational flow

Circulation:

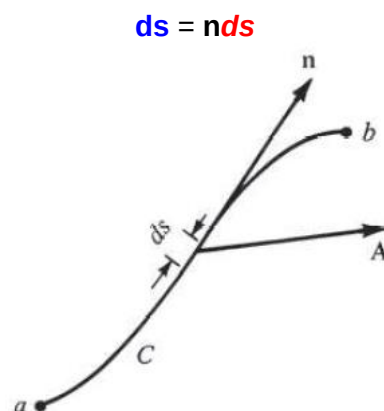
In physics, circulation is **the line integral of a vector field around a closed curve**. In fluid dynamics, the field is the fluid velocity field.

Line integrals, by mathematical convention the positive sense of the line integral is counter clockwise. However, In aerodynamics, it is convenient to consider a positive circulation as being clockwise.

Consider a closed curve C in a flow field. Each curve segment $d\mathbf{s}$ are a vector and has an associated velocity vector with it.



Let $d\mathbf{s}$ be an elemental length of the curve, and $\mathbf{n}$ be a unit vector tangent to the curve. The vector $d\mathbf{s}$ is

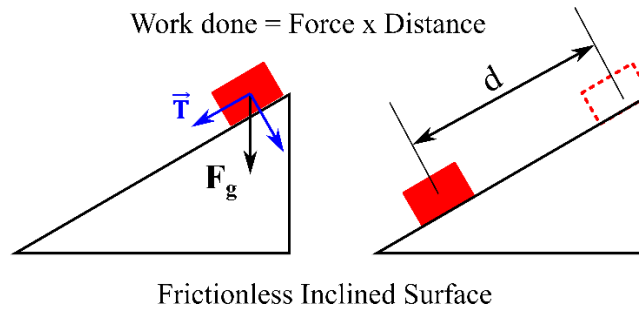


Amount of flow around the loop or total flow around the loop is

$$\Gamma = \oint_C \vec{V} \cdot d\vec{s}$$

Where $d\vec{s}$ is the vector of the segment of a curve associated with elemental length of the curve ds and a unit vector $\mathbf{n}$ tangent to the curve.

Example:



Work done by the gravity field (vector field) on the box placed on frictionless surface can be written as,

$$W = (\vec{F} \cdot \vec{T})d$$

$\vec{T}$ is the tangential vector shows the tangential direction. $(\vec{F} \cdot \vec{T})$ proportion of the force in the tangential direction.

Similarly, the work done by the vector field to move the fluid elements along the curve can be defined as,

$$W = \Gamma = \oint_C (\vec{V} \cdot \vec{n})ds$$

We know that $\vec{n} = \frac{d\vec{s}}{ds}$, Hence the circulation can be written as,

$$\Gamma = \oint_C \vec{V} \cdot d\vec{s}$$

References:

1. Anderson, J. (2011): *Fundamentals of Aerodynamics (SI units)*. McGraw hill.
2. Cengel, Y., & Cimbala, J. (2013): *Fluid mechanics fundamentals and applications (SI units)*. McGraw Hill.